An Information-Theoretic View of Cache-Aided Networks: Part 2 – Caching for Erasure Broadcast Channels

Michèle Wigger

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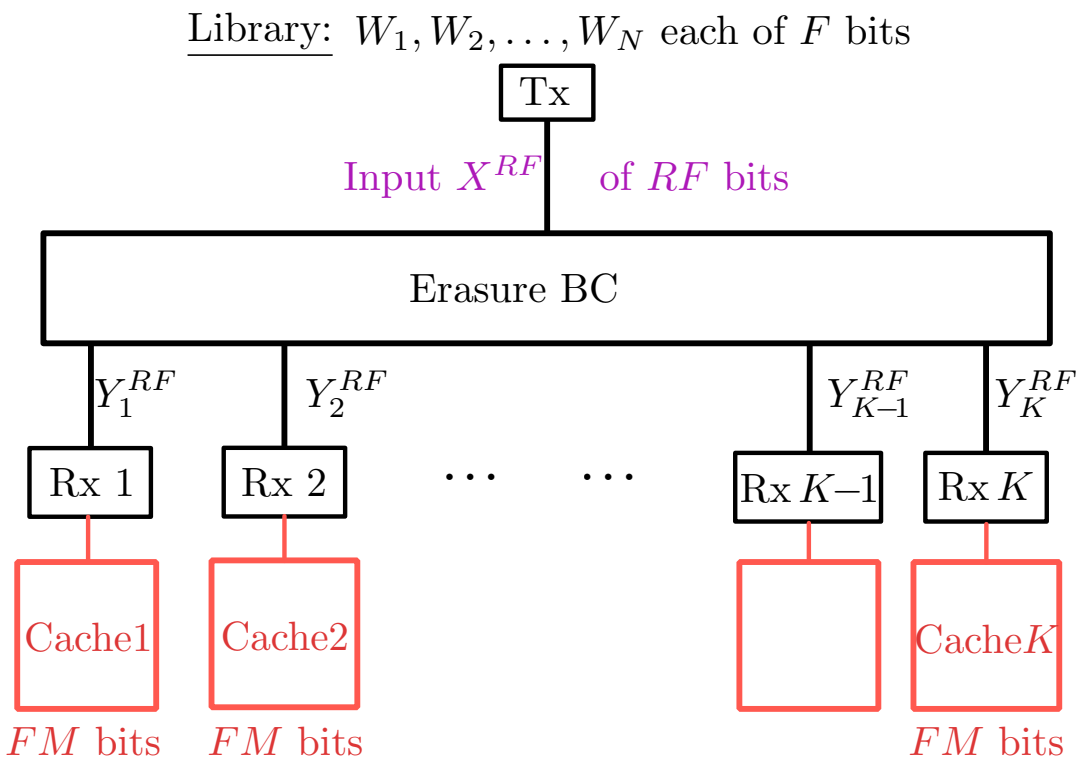
Delivery over Erasure Broadcast Channel (BC), Homework Exercise 1



- Each transmitted bit erased at each receiver with probability $\delta > 0$, irrespective of all other bits
- Adapt the coded caching upper bound on $R^\star(M)$ to this setup!

Noisy Setup Requires Vanishing Error Probability

- Cache placement $\quad Z_k = g_k(W_1, \ldots, W_N) \quad$ consists of FM bits
- Delivery encoding $\quad X^{RF} = f(W_1, \ldots, W_N, d_1, \ldots, d_K) \quad$ of FR bits
- Delivery decoding $\quad \hat{W}_k = \varphi_k(Y^{RF}, Z_k, d_1, \ldots, d_K)$.

- Goal: $\quad \Pr\left[\hat{W}_k = W_{d_k}\right] \to 0 \quad$ as $F \to \infty$ for all $k = 1, \ldots, K$

  $\to$ Need to tolerate errors because the channel has nonzero probability of experiencing lots of erasures.

Rate-Memory Tradeoff for Erasure Broadcast Channels ($N \geq K$)

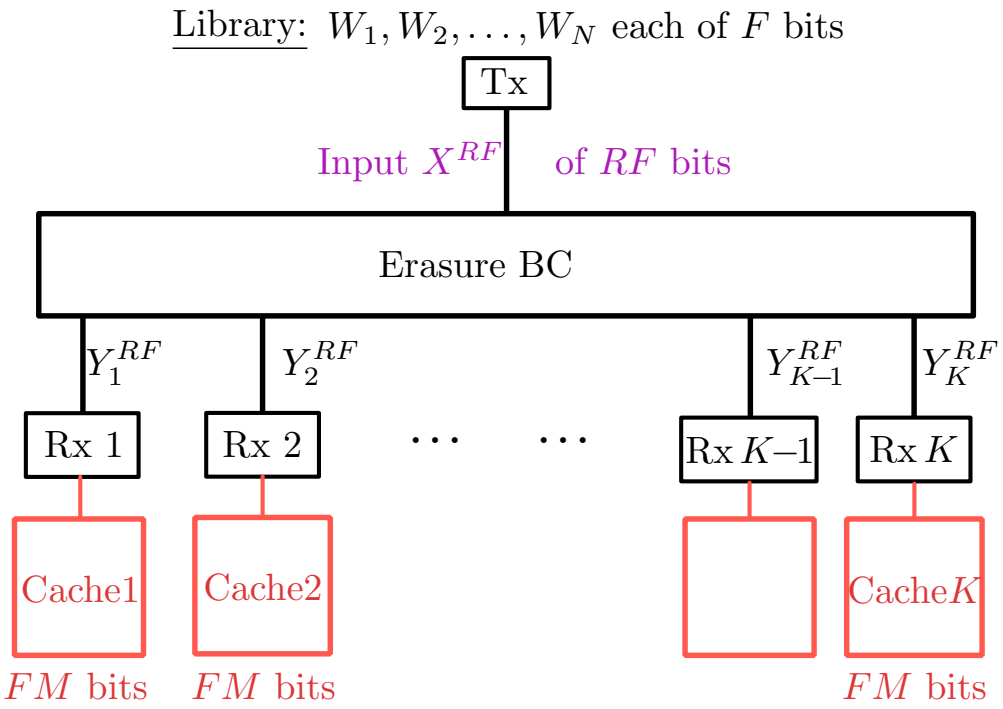
- Capacity of a erasure broadcast channel (EBC) is $(1-\delta)$, both for sending common messages and private messages.
- That means, by sending FR inputs, we can convey $FR(1-\delta)$ information bits with arbitrarily small probability of error as $F \to \infty$.
- So now each XOR packet requires $(1-\delta)$ times more transmission slots to be setn than before. As a consequence:

$$R^\star(M) \leq \text{convhull}\Big\{(M_t, R_t)\colon\ t = 0, 1, \ldots, K\Big\}$$

where

$$\begin{aligned} M_t &= \frac{tN}{K} \\ R_t &= \frac{1-\delta}{K}\Big(1 - \frac{M_t}{N}\Big)\frac{1}{1+\frac{KM_t}{N}}. \end{aligned}$$

Delivery over Erasure BCs, Homework Exercise 2



- Erasure probability at receivers $1, \ldots, K_w$ is δ_1, where $K_w < K$
- Erasure probability at receivers $K_w + 1, \ldots, K$ is $\delta_2 < \delta_1$
- Adapt the coded caching upper bound on $R^\star(M)$ to this setup!

Rate-Memory Tradeoff for Asymmetric EBCs

- The XOR packets that are meant for at least one weak receiver require $(1-\delta_1)^{-1}$ times more channel inputs than over the noisefree channel
- The XOR packets that are meant for none of the strong receivers require $(1-\delta_2)^{-1}$ times more channel inputs than over the noisefree channel
- $\binom{K_s}{t+1}$ XOR packets intended for only the strong receivers, where t denotes the parameter of the coded caching scheme
- $\binom{K}{t+1} - \binom{K_s}{t+1}$ of the XOR packets intended for at least one weak receiver

EBCs with Unequal Channel Strengths

- In this coded caching algorithm requires:

$$R^\star(M) \leq \text{convhull}\Big\{(M_t, R_t)\colon\ t = 0, 1, \ldots, N\Big\}$$

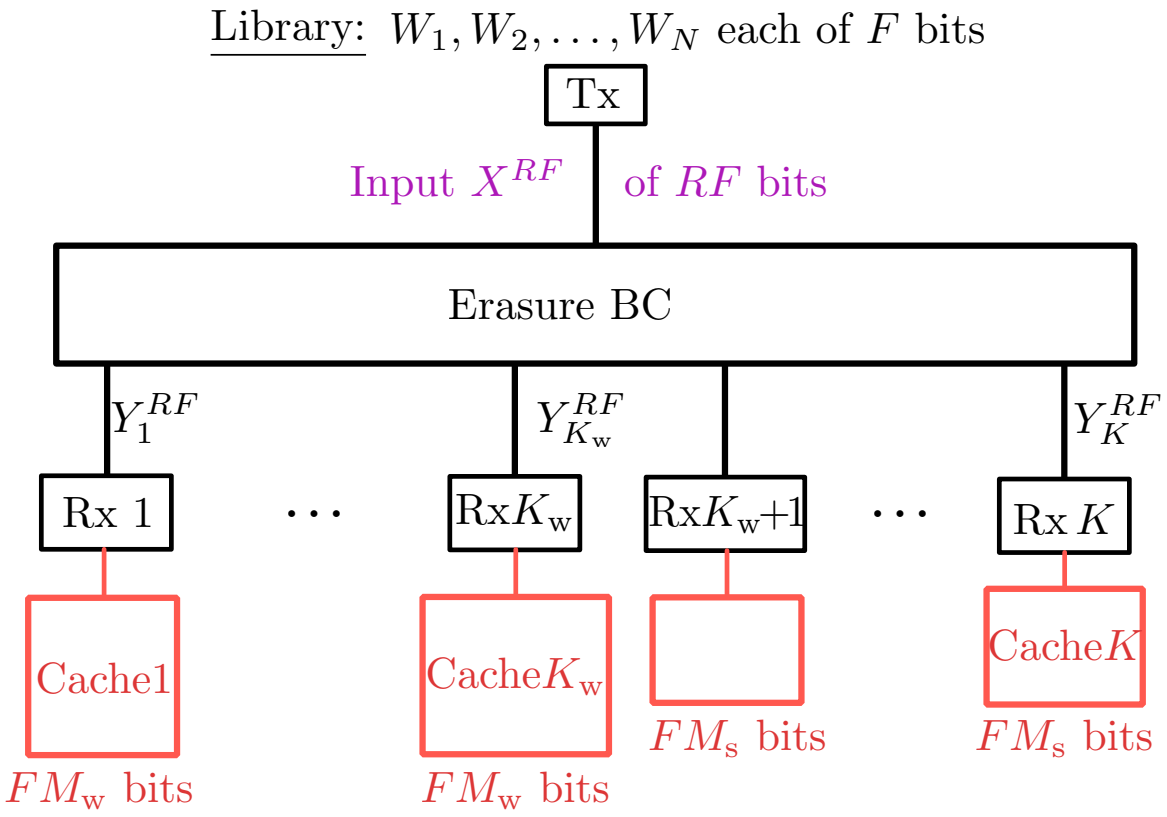
where

$$\begin{aligned} M_t &= \frac{tN}{K} \\ R_t &= \frac{K}{(1-\delta_2)}\Big(1 - \frac{M_t}{N}\Big)\frac{1}{1+\frac{KM_t}{N}} + \underbrace{\frac{\binom{K}{t+1} - \binom{K-K_w}{t+1}}{\binom{K}{t}}\left(\frac{1}{1-\delta_1} - \frac{1}{1-\delta_2}\right)}_{\text{penalty caused by weak receivers}} \end{aligned}$$

Penalty Caused By Weak Receivers

- Weak receivers cause rate-penalty!
- As we will see, this holds only for asymmetric situations
- Penalty caused by weak receivers can partially be removed in asymmetric setups where weak receivers need less information or when they have larger cache memories
- Efficient elimination of rate penalty requires new coding approach!

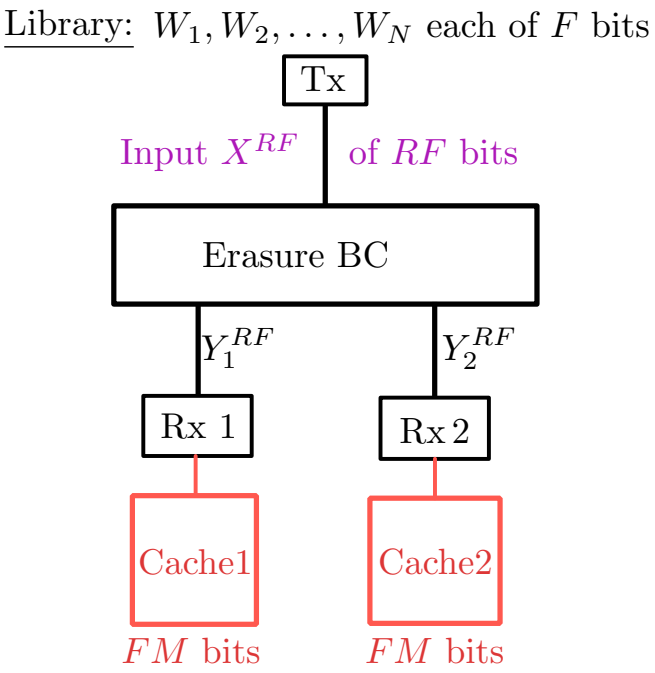
Can Cache Assignment Resolve Penalty?



- Move part of the cache memories from strong receivers to weak receivers
  $\rightarrow$ Idea: Help more the weak receivers to make the network more balanced

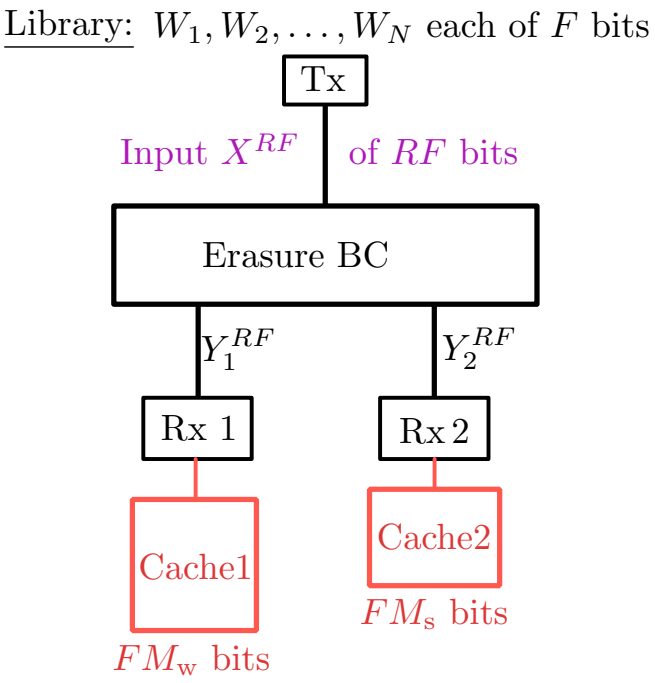
- How to exploit additional cache memories?
  $\rightarrow$ Coded caching only works with equal cache memories at all receivers

Two-User Example with Symmetric Cache Memories



- Coded caching with parameter $t = 1$:
  - Split each W_d into $(W_d^{(1)}, W_d^{(2)})$ of size $F/2$ bits
  - Placement: Rx 1 caches $\{W_d^{(1)}\}$ and Rx 2 caches $\{W_d^{(2)}\}$ $\rightarrow$ cache memory $M = 1/2$
  - Delivery: Encode $W_{d_1}^{(2)} \oplus W_{d_2}^{(1)}$ using a capacity-achieving code
- For $p(error) \rightarrow 0$ as $F \rightarrow \infty$, need $R \geq \frac{1}{2(1-\delta_1)}$

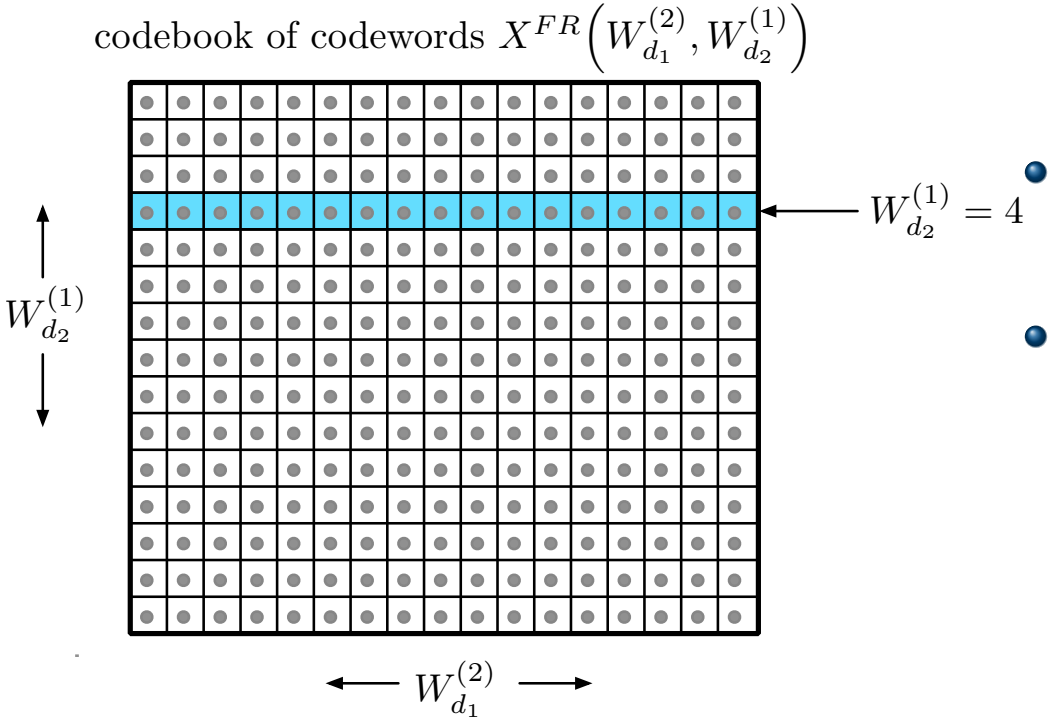
Two-User Example with Asymmetric Cache Memories



- Assign cache memory inverse proportionally to channel capacities: $M_{\mathsf{w}} = \frac{1-\delta_2}{2-\delta_1-\delta_2} N$ and $M_{\mathsf{s}} = \frac{1-\delta_1}{2-\delta_1-\delta_2} N$
- Split each $W_d = (W_d^{(1)}, W_d^{(2)})$ with sizes $\frac{1-\delta_2}{2-\delta_1-\delta_2} F$ and $\frac{1-\delta_1}{2-\delta_1-\delta_2} F$ bits
- Placement: store $\left\{W_d^{(1)}\right\}_{d=1}^{N}$ in Cache 1 and $\left\{W_d^{(2)}\right\}_{d=1}^{N}$ in Cache 2

Use a "Piggyback-Code" to Delivery $W_{d_1}^{(2)}$ and $W_{d_2}^{(1)}$

Randomly generate all codewords IID by choosing all entries IID.



- Rx 1 knows $W_{d_2}^{(1)}$
  $\rightarrow$ restricts decoding to a row
- Rx 2 knows $W_{d_1}^{(2)}$
  $\rightarrow$ restricts decoding to a column

- For Rx 1 to be able to decode, X^{FR} needs to be of size $\frac{F}{2-\delta_1-\delta_2}$ bits
- For Rx 2 to be able to decode, X^{FR} needs to be of size $\frac{F}{2-\delta_1-\delta_2}$ bits

"Piggyback-Code" removes Penalty caused by Weak Receiver

For $p(error) \to 0$ as $n \to \infty$ we need $R \geq \dfrac{1}{2 - \delta_1 - \delta_2}$

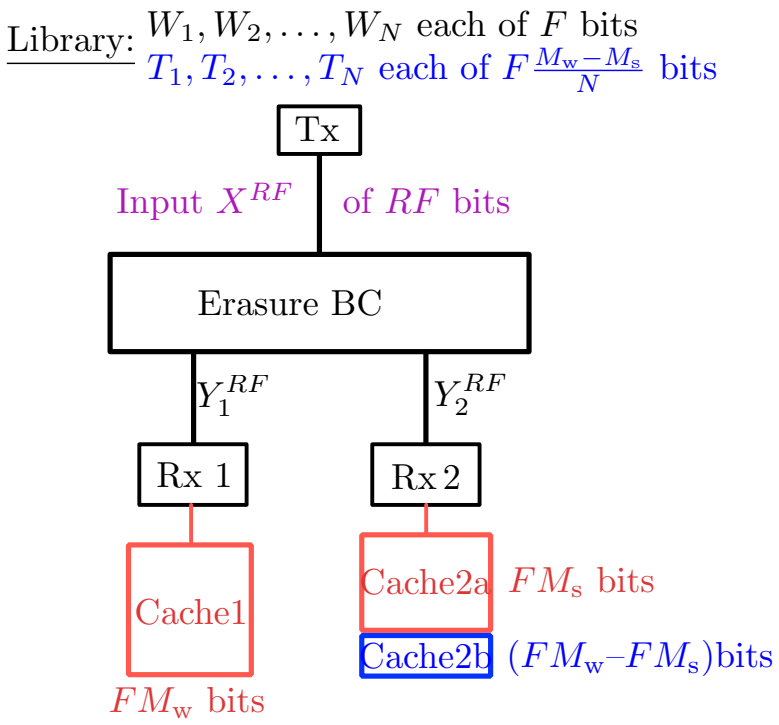
- Same performance as if only one of the receivers was present!
- Weaker receiver does not penalize stronger receiver!

Piggyback-Coding Extends to K Receivers

- Piggyback coding extends to arbitrary number of receivers and different erasure probabilities
- Extends to general degraded broadcast channels (BC)
- Modify coded caching as follows:
  - Size of a subpart of files depends on channel strengths of the receivers caching this subpart
  - Choose $t+1$-dimensional piggyback codebook for delivery communication to each subset of $t+1$ receivers

Size of subparts (and thus of cache memories) is chosen so that each piggyback codebook is decoded using the same time by each of the involved receivers

Higher Resolution At Stronger Receivers



- Let $T_1, \ldots, T_N$ be higher resolution info. required at Rx 2, not at Rx 1
- Store all $T_1, \ldots, T_N$ in Rx 2's cache memory
- Apply placement and delivery strategies described before